

The REPACT Tool

User Manual



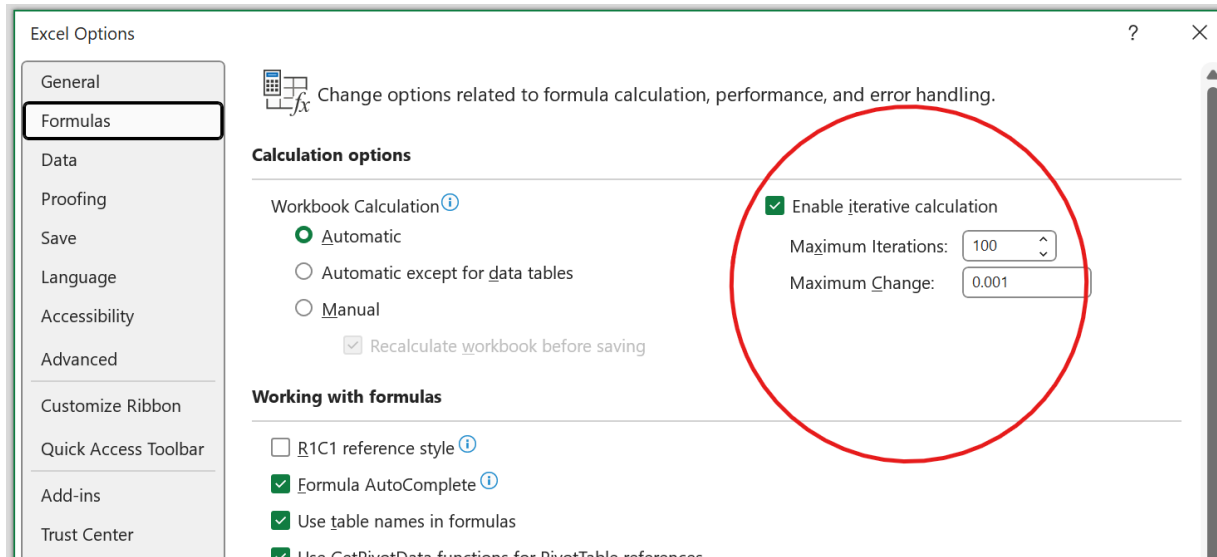
September 30, 2024



- Reuse of Existing Pipelines for Adapted Carbon Transport (REPACT) is an Excel-based screening tool.
- This tool enables the user to determine whether a pipeline **originally deployed** for natural gas transport, can be **reused** for CO₂ transport.
- This manual will walk you through an intended usage statement, setup instructions, a list of inputs and outputs in the tool, and frequently used protocols.
- **Please note that this is a first-pass screening tool. The approach presented within this tool is valid for high-level evaluation only and should not be used to make engineering, financial, and/or regulatory decisions.**

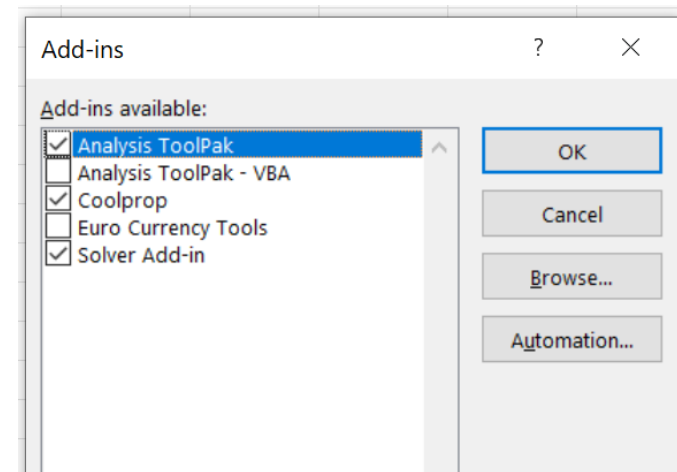
- The REPACT tool is designed as a “screening model” to guide the user on whether a retiring/retired natural gas pipeline (the candidate pipeline) can be considered for CO₂ transport by,
 - a) Taking the user through a set of pre-screening regulatory questions on the candidate pipeline
 - b) Providing potential phases in which CO₂ transport may be considered, if the candidate pipeline passes the regulatory pre-screening questions
 - c) Providing estimates of CO₂ flow rates and outlet pressure (for gaseous and supercritical phases) that may be supported by the candidate pipeline, given its physical characteristics, inlet pressure and temperature
 - d) Providing the option to enable intermediate pressure boosting to meet minimum pressure requirements
 - e) Providing sensitivities to assist user with permissible ranges on flow rates and inlet pressures that the candidate pipeline can support
- The intended audience for this tool includes pipeline operators, CCS project developers, researchers (CO₂ transport pathways), change agents in energy communities (repurposing existing energy assets).

- The tool uses macros so the user would need to “Enable Content” each time the tool is opened.
- A circular reference is built into the tool to solve for outlet pressures when the user does not wish to input a value. It is essential that the user “Enable(s) iterative calculation” using Excel Options (see below).



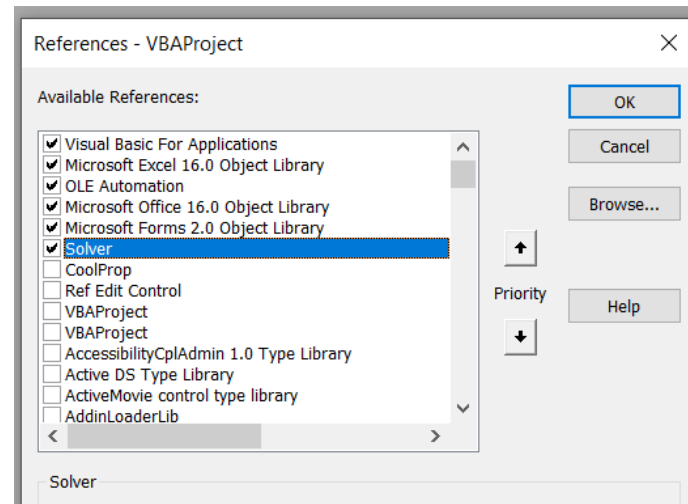
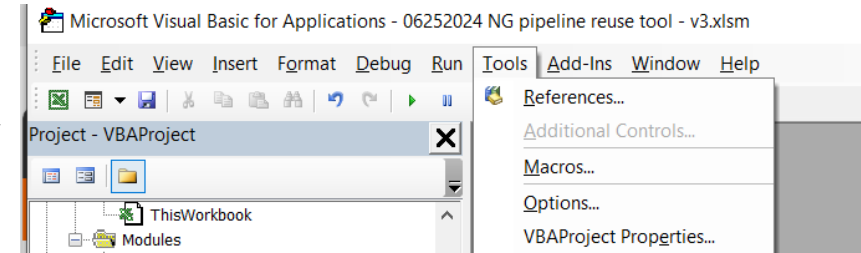
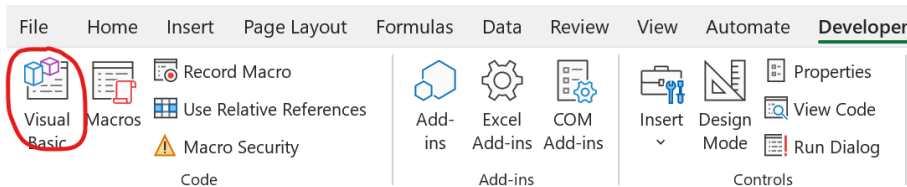
Excel Setup (contd.)

- This tool contains buttons that rely on Excel's "Solver" functionality. Please ensure that the Solver Add-In is enabled by following these steps.
- Go to File > Options > Add-Ins > Click "Go" next to "Manage:" > Ensure Solver Add-in is enabled > Click OK.



Excel Setup (contd.)

- The user would need to ensure that the macros can work correctly with the Solver. For this, go to “Developer” on the menu bar and open Visual Basic. Go to Tools and select References. Ensure the box next to Solver is checked and click OK.



Tool assumptions

- REPACT follows the assumptions mentioned below.
 - Single phase flow (with inbuilt checks for phase change)
 - No elevation change assumed
 - Isothermal flow
 - No heat transfer calculations
 - Inlet and outlet pressure loss is not included
 - Properties are based on pure CO₂

Tool Inputs and Outputs

General inputs/outputs

- All tool input cells have a blue background and output cells have a golden background. Cells with a white background have fixed values.

Legend
Blue cell: input
Gold cell: output
Cell with no fill: fixed values

Tab Name	Descriptor Cell/Sub-Category	Cell Type and Value	Notes
Pre-feasibility screening	Nominal diameter (inches)	Numeric input from dropdown menu	Pipeline diameter
	Material	Input from dropdown menu	Pipeline material selection
	Grade	Input from dropdown menu	Pipeline grade selection
	Roughness (inches)	Numeric input	Default value 0.0018; more options available in a user note
	Length (miles)	Numeric input	Pipeline length
	Sub-Category: {Regulatory screening questions}	All are Yes/No/Unsure inputs from dropdown menu	Remarks column provides impact of chosen input

Tools Inputs and Outputs (contd.)

General inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Pre-feasibility screening	Are pipeline materials suitable for CO ₂ transport?	Output cell with a Yes/No/Unlikely value	Remarks column provides details on the output
	Can the existing pipeline be used for CO ₂ transport?	Output cell with a Yes/No/Unlikely values	Interpretation column provides details on the output
	Which phase(s) could the CO ₂ be transported in?	CO ₂ phase names	Potential phases in which the pipeline could support transport

Tools Inputs and Outputs (contd.)

Gaseous phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Gaseous phase transport	Inlet pressure (psig)	Numeric input	Permissible range is 45-850 psig. Remarks column shows pressure in Megapascals
	Inlet temperature (°F)	Numeric input	Permissible range is 21-94 °F. Remarks column shows temperature in °C.
	Predicted CO ₂ phase at inlet conditions	Text output	Fluid phase predicted by the tool
	Do you have a desired outlet pressure?	Yes/No input	User can choose whether to provide a desired outlet pressure.
	Insert desired outlet pressure (psig) OR This input is skipped	Numeric input (must be less than or equal to inlet pressure)	This input is considered only if the input to "Do you have a desired outlet pressure?" is "Yes".

Tools Inputs and Outputs (contd.)

Gaseous phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Gaseous phase transport	Calculated/Desired outlet pressure (psig)	Numeric output	Desired outlet pressure (calculated by the tool in case of no user input).
	Target flow rate (tonnes/year)	Numeric Input	Remarks column shows value in billion standard cubic feet/year
	Average Velocity (m/s)	Numeric output	Average velocity based on flow characteristics and CO ₂ density
	Resultant outlet pressure without booster stations (psig)	Numeric output	Outlet pressure based on inlet temperature and pressure conditions, and pipeline characteristics, before boosting.
	Predicted CO ₂ phase at outlet conditions	Text output	Fluid phase predicted by the tool. Remarks column let's the user know if there is a phase change within the pipeline.
	Is boosting required at this flow rate?	Yes/No output	"Yes" when the resultant outlet pressure is less than max{45psig, calculated outlet pressure}

Tools Inputs and Outputs (contd.)

Gaseous phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Gaseous phase transport	Allow intermediate booster stations?	Yes/No input from dropdown menu	User can choose to allow intermediate boosting via this input
		“Find boosters” button (input)	Clicking this button will solve for up to two intermediate boosters to increase the “calculated” outlet pressure to at least the “desired” outlet pressure level.
	Resultant outlet pressure with booster stations (psig) (visible only when boosting is enabled)	Numeric output	Outlet pressure after enabling and solving for intermediate pressure boosting. Valid after “Find boosters” is clicked.
	Number of booster stations required to maintain minimum pressure of 45 psi	Output in set {'-', 0, 1, 2, 2+}	Valid after “Find boosters” is clicked.
	At what flow rate would the outlet pressure drop to 45 psig?	“Reset flow rate” button (input)	Resets the value of “Target flow rate” input to 10,000 tonnes/year.
		“Maximize flow rate” button (input)	Maximizes the “Target flow rate” while keeping the pipeline pressure above 45 psig.

Tools Inputs and Outputs (contd.)

Gaseous phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Gaseous phase transport	Average density of the fluid across the pipeline (kg/m ³)	Numeric output	Average of pipeline's inlet and outlet temperature and pressure values, valid for gaseous flow only.
	Average of dynamic viscosity (μ) across the pipeline (Pa s)	Numeric output	Average of pipeline's inlet and outlet temperature and pressure values, valid for gaseous flow only.

Tools Inputs and Outputs (contd.)

Supercritical phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Supercritical phase transport	Inlet pressure (psig)	Numeric input	Permissible range is 1100-1800 psig. Remarks column shows pressure in Megapascals
	Inlet temperature (°F)	Numeric input	Permissible range is 96-140 °F. Remarks column shows temperature in °C.
	Predicted CO ₂ phase at inlet conditions	Text output	Fluid phase predicted by the tool
	Do you have a desired outlet pressure?	Yes/No input	User can choose whether to provide a desired outlet pressure.
	Insert desired outlet pressure (psig) OR This input is skipped	Numeric input (must be less than or equal to inlet pressure)	This input is considered only if the input to "Do you have a desired outlet pressure?" is "Yes".

Tools Inputs and Outputs (contd.)

Supercritical phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Supercritical phase transport	Calculated/Desired outlet pressure (psig)	Numeric output	Desired outlet pressure (calculated by the tool in case of no user input).
	Target flow rate (tonnes/year)	Numeric Input	Remarks column shows value in billion standard cubic feet/year
	Average Velocity (m/s)	Numeric output	Average velocity based on flow characteristics and CO ₂ density
	Resultant outlet pressure without booster stations (psig)	Numeric output	Outlet pressure based on inlet temperature and pressure conditions, and pipeline characteristics, before boosting.
	Predicted CO ₂ phase at outlet conditions	Text output	Fluid phase predicted by the tool. Remarks column let's the user know if there is a phase change within the pipeline.
	Is boosting required at this flow rate?	Yes/No output	"Yes" when the resultant outlet pressure is less than max{1100 psig, calculated outlet pressure}

Tools Inputs and Outputs (contd.)

Supercritical phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Supercritical phase transport	Allow intermediate booster stations?	Yes/No input from dropdown menu	User can choose to allow intermediate boosting via this input
		"Find boosters" button (input)	Clicking this button will solve for up to two intermediate boosters to increase the "calculated" outlet pressure to at least the "desired" outlet pressure level.
	Resultant outlet pressure with booster stations (psig) (visible only when boosting is enabled)	Numeric output	Outlet pressure after enabling and solving for intermediate pressure boosting. Valid after "Find boosters" is clicked.
	Number of booster stations required to maintain minimum pressure of 1100 psi	Output in set {'-', 0, 1, 2, 2+}	Valid after "Find boosters" is clicked.
	At what flow rate would the outlet pressure drop to 1100 psig?	"Reset flow rate" button (input)	Resets the value of "Target flow rate" input to 10,000 tonnes/year.
		"Maximize flow rate" button (input)	Maximizes the "Target flow rate" while keeping the pipeline pressure above 1100 psig.

Tools Inputs and Outputs (contd.)

Supercritical phase inputs/outputs

Tab Name	Descriptor Cell	Cell Type and Value	Notes
Supercritical phase transport	Average density of the fluid across the pipeline (kg/m ³)	Numeric output	Average of pipeline's inlet and outlet temperature and pressure values, valid for supercritical flow only.
	Average of dynamic viscosity (μ) across the pipeline (Pa s)	Numeric output	Average of pipeline's inlet and outlet temperature and pressure values, valid for supercritical flow only.

1. Reset flow rate

This button click resets the value of target flow rate to 10,000 tonnes/year; this is a good value to

- i. Have a starting point for trying out multiple flow rates,
- ii. Reset occasional erroneous tool output at edge cases for temperatures and pressures.

2. Reset flow rate + Maximize flow rate

This series of button clicks can be used to first reset the flow rate to 10,000 tonnes/year and then find the maximum flow rate that can be supported by the candidate pipeline,

- i. At the given inlet temperature and pressure,
- ii. While staying above the minimum permissible pressure (45 psig for gaseous and 1100 psig for supercritical) at all points in the pipeline

3. Allow intermediate booster stations? “Yes” + Find boosters

These set of inputs are useful when pressure boosting is required to attain target flow rate (the answer to the cell “Is boosting required at this flow rate?” is “Yes”). Upon clicking the “Find boosters” button, Excel Solver estimates the number of intermediate pumps required to attain desired outlet pressure at the end of the pipeline.

Please note that it is essential to *click on* the “Find boosters” button every time an input (for instance the target flow rate) is changed, because the button acts as a trigger to the Excel Solver.

4. Referring the Phase Diagram tab

This tab contains diagrams to assist the user in identifying good ranges for inlet temperatures and pressures, depending on the CO₂ phase. The diagram titled “Current input/output conditions” plots 4 points (2 red points for gaseous and 2 purple points for supercritical tabs) to ensure that there is no phase change between inlet and outlet conditions.

Continuity with other NETL tools

- The calculations of the REPACT tool are available for gaseous and supercritical flows.
- Other NETL tools that exist for CO₂ transport only model new infrastructure. This is different from REPACT in that it uses existing natural gas infrastructure.
- REPACT follows the assumptions like –
 - Single phase flow (with inbuilt checks for phase change)
 - No elevation change assumed
 - Isothermal flow
 - No heat transfer calculations
 - Inlet and outlet pressure loss is not included
 - Properties are based on pure CO₂
- Since REPACT is a high-level screening model, it does not currently tie to other higher-fidelity tools. However, upcoming NETL screening tools for CO₂ buffer storage (BUCS) and maritime transport are expected to be linked well to REPACT.

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